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(54) PROCESS FOR THE PREPARATION OF PLEUROMUTILINS

VERFAHREN ZUR HERSTELLUNG VON PLEUROMUTILINEN

PROCÉDÉ DE PRÉPARATION DE PLEUROMUTILINES

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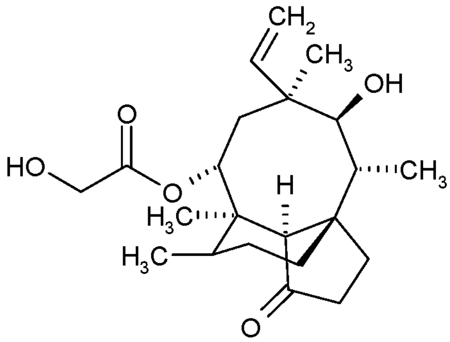
Description

[0001] The present invention relates to a process for the preparation of crystalline 14-O-{[(4-amino-2-hydroxy-cyclohexyl)-sulfanyl]-acetyl}-mutilin and crystalline salts thereof.

[0002] Pleuromutilin, a compound of formula



Pleuromutilin

is a naturally occurring antibiotic, e.g. produced by the basidiomycetes Pleurotus mutilus and P. passeckerianus, see e.g. The Merck Index, 12th edition, item 7694.

[0003] A number of further pleuromutilins having the principle ring structure of pleuromutilin and being substituted at the primary hydroxy group have been developed, e.g. as antimicrobials. Due to their pronounced antimicrobial activity, a group of pleuromutilin derivatives, amino-hydroxy-substituted cyclohexylsulfanylacetylmutilins, as disclosed in WO 2008/113089, have been found to be of particular interest. As described in WO2008/11089 14-O-{[(4-Amino-2-hydroxy-cyclohexyl)-sulfanyl]-acetyl}-mutilins are particularly useful compounds because they demonstrate activity against Gram-positive and Gram-negative pathogens e.g. associated with respiratory tract and skin and skin structure infections. For the production of substantially pure isomers/diastereomers of this group of compounds, there is a need for a production process which is convenient for use on an industrial scale and which also avoids the use of costly starting materials, environmentally hazardous reagents and solvents or time consuming and laborious purification steps. The production process described in WO 2008/113089 involves chromatographic purification of the compounds prepared according to individual synthesis steps and the final diastereomers are separated by chiral HPLC chromatography which cannot be used on industrial scale.

[0004] Surprisingly, crystalline intermediates have been found which on the one hand have unexpected chemical purification potential which is important for the production processes for pure amino-hydroxy-substituted cyclohexylsulfanylacetylmutilins avoiding chromatographic purification and separation steps.

[0005] It has to be pointed out that 14-O-{[(4-amino-2-hydroxy-cyclohexyl)-sulfanyl]-acetyl}-mutilins are potential new drug substances for the human market with regulatory requirements defined in the corresponding ICH guidelines (International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use). The ICH guideline on impurities in new drug substances (Q3A(R2)) includes the following thresholds:

Maximium daily dose	Reporting threshold	Identification threshold	Qualification threshold
≤ 2g	0.05%	0.10%	0.15%
> 2g	0.03%	0.05%	0.05%

[0006] As can be seen from the ICH thresholds above it is desirable to have all individual unknown impurities below 0.10% area and the structure elucidated impurities below 0.15%, respectively. The process provided according to the present invention enables to produce APIs (Active Pharmaceutical Ingredients) within the desired specifications and fulfilling ICH requirements.

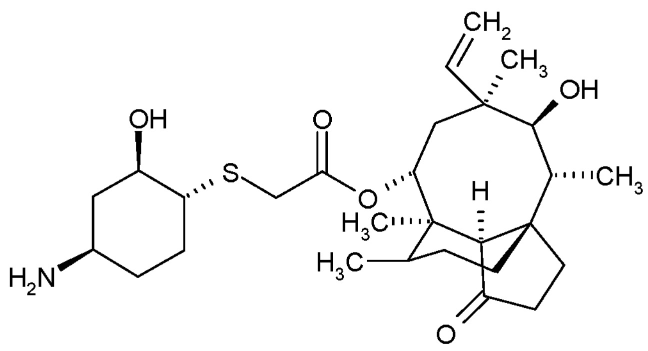
[0007] On the other hand, even more surprisingly, the crystalline intermediates yields to significant chiral enrichment which has a huge benefit in the production of the pure stereoisomers starting from cheaper racemic materials or less chirally pure starting materials. The described processes do not involve any chromatographic purification neither normal nor chiral phase in contrast to the synthetic procedures described in WO2008/113089 wherein is disclosed e.g. in Example 1, Step B that 14-O-{[(4-amino-2-hydroxy-cyclohexyl)-sulfanyl]-acetyl}-mutilins was isolated in the form of diastereomeric mixtures as colorless amorphous foams after normal phase chromatography. The chiral pure diastereomers are described to have been received in WO2008/113089, e.g. in Example 1A after subjecting the mixture to chiral chromatography whereafter the separated pure diastereomers were isolated in the form of colorless amorphous foams.

[0008] Chiral chromatography, however is not a technology which can be applied on industrial large scale, and moreover no crystalline salts of 14-O-{[(4-amino-2-hydroxy-cyclohexyl)-sulfanyl]-acetyl}-mutilins were obtained according to WO2008/113089.

[0009] In contrast to that, crystalline pharmaceutical acceptable salts of 14-O-{[(4-amino-2-hydroxy-cyclohexyl)-sulfanyl]-acetyl}-mutilins having surprising and superior properties over the amorphic prior art salts disclosed in WO2008/113089 have been found; e.g. surprisingly the chemical stability of the crystalline salts disclosed herein is improved over the amorphic salt forms; and also and in addition the crystalline salts disclosed herein show a surprising low hygroscopicity.

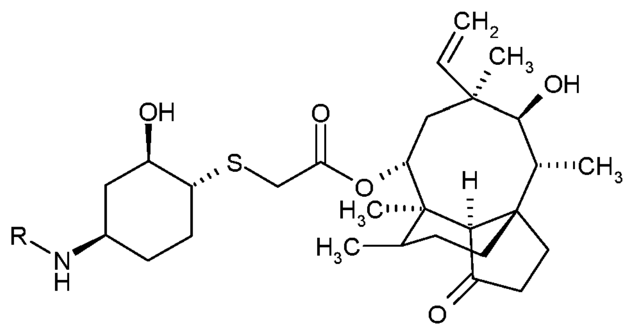
[0010] Processes for the preparation of such crystalline salts wherein the salts may be obtained in a single stereoisomeric form from 14-O-{[(4-amino-2-hydroxy-cyclohexyl)-sulfanyl]-acetyl}-mutilins and processes for the preparation of stereoisomerically pure 14-O-{[(4-amino-2-hydroxy-cyclohexyl)-sulfanyl]-acetyl}-mutilins in crystalline form as a basis for the crystalline salts have also been found.

[0011] The present invention provides a process for the preparation of a compound of formula I



I

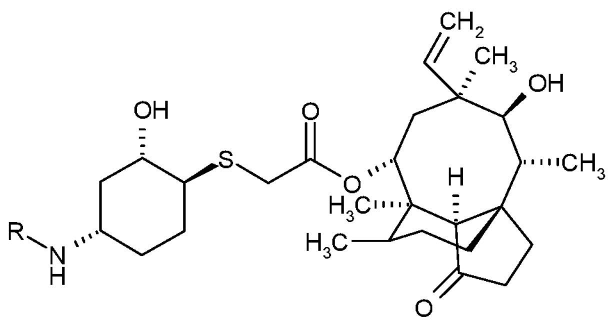
in the form of a single stereoisomer in crystalline form, comprising deprotecting the amine group either in a compound of formula IIa



IIa

,

or in a mixture of a compound of formula IIa with a compound of formula IIb



IIb

wherein R is an amine protecting group, and isolating a compound of formula I obtained in the form of a single diastereomer in crystalline form either directly from the reaction mixture or via recrystallization in organic solvent, whereby isolating the compound of formula I in the form of a single diastereomer in crystalline form comprises one of the following steps:

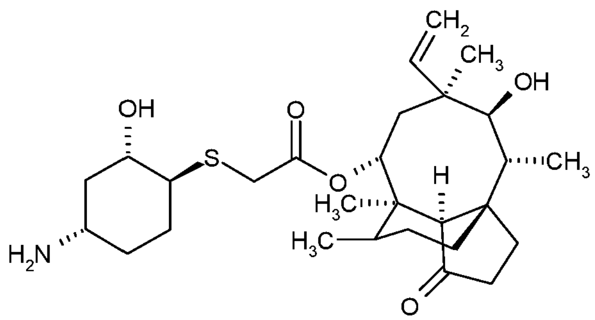
a) treating a solution of a compound of formula I in a single stereoisomeric form obtained in the polar organic solvent CH_2Cl_2, with an antisolvent selected from diisopropylether (DIPE) or tert-butyl methyl ether (MTBE), preferably DIPE
b) treating a solution of a compound of formula I in a single stereoisomeric form obtained in the polar organic solvent CH_2Cl_2, after concentration with an alcohol, such as *n*-butanol;
c) taking up an isolated crude material of a compound of formula I in a single stereoisomeric form, in an ether, e.g. tetrahydrofuran (THF), and optionally treating it with an antisolvent selected from DIPE or MTBE;
d) obtaining a compound of formula I in the form of a single diastereomer in crystalline form via (re-)crystallization in an organic solvent, wherein the organic solvent is an alcohol, such as n-butanol.

[0012] The dependent claims set out preferred embodiments.

[0013] Disclosed herein is a compound of formula I as defined above in the form of a single stereoisomer in crystalline form.

[0014] Disclosed herein is also a compound of formula IIa.

[0015] In a compound of formula I, or IIa, respectively, the carbon atoms of the cyclohexyl ring to which the hydroxy group, the amine group and the sulfanyl-acetyl-mutilin group are attached are all in the R configuration and thus a compound of formula I, or IIa represents an optionally amine protected 14-O-{[(1*R*,2*R*,4*R*)-4-amino-2-hydroxy-cyclohexylsulfanyl]-acetyl}-mutilin. In contrast to that, in a compound of formula Ib



Ib

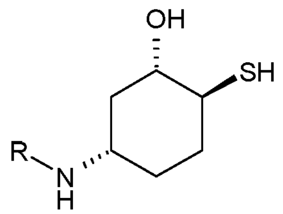
or IIb the carbon atoms of the cyclohexyl ring to which the hydroxy group, the amine group and the sulfanyl-acetyl-mutilin group are attached are all in the S configuration and thus a compound of formula IIb represents an optionally amino protected 14-O-{[(1*S*,2*S*,4*S*)-4-Amino-2-hydroxy-cyclohexylsulfanyl]-acetyl}-mutilin.

[0016] An amine protecting group includes protecting groups known to a skilled person and which are removable under acidic, basic, hydrogenating, oxidative or reductive methods, e.g. by hydrogenolysis, treatment with an acid, a base, a hydride, a sulfide. Appropriate amine protecting groups e.g. are described in T. W. Greene, P. G. M. Wuts, Protective Groups in Organic Synthesis, Wiley-Interscience, 4th edition, 2007, particularly p. 696-868.

[0017] Amine protecting groups which may be conveniently used in a process according to the present invention include for example

- benzyloxycarbonyl (Cbz), removable e.g. by hydrogenolysis,
- *p*-methoxybenzyl carbonyl (Moz or MeOZ), removable e.g. by hydrogenolysis,
- *tert*-butyloxycarbonyl (BOC), removable e.g. by treatment with a strong acid, such as HCl, H_3PO_4 or CF_3COOH,
- trifluoroacetyl, removable e.g. by treatment with a base, such as NaOH, K_2CO_3, Cs_2CO_3,
- 9-fluorenylmethyloxycarbonyl (FMOC), removable e.g. by treatment with a base, such as piperidine,
- benzyl (Bn), removable e.g. by hydrogenolysis;
- *p*-methoxybenzyl (PMB), removable e.g. by hydrogenolysis;
- 3,4-dimethoxybenzyl (DMPM), removable e.g. by hydrogenolysis;
- *p*-methoxyphenyl (PMP), removable e.g. by treatment with ammonium cerium(IV) nitrate (CAN),
- tosyl (Tos), removable e.g. by treatment with concentrated acid, such as HBr, H_2SO_4, or by treatment with strong reducing agents, such as sodium in liquid ammonia, sodium naphthalene,
- groups which form with the amine sulfonamides other than Tos-amides, e.g. including 2-nitrobenzenesulfonamide (nosyl) or o-nitrophenylsulfenyl (Nps), removable e.g.by treatment with samarium iodide, tributyltin hydride,
- benzylidene, removable e.g. by treatment with trifluoromethanesulfonic acid, trifluoroacetic acid, dimethyl sulfide;
- triphenylmethyl (trityl, Tr), dimethoxytrityl (DMT), e.g. removable by treatment with an acid, such as trifluoroacetic acid;

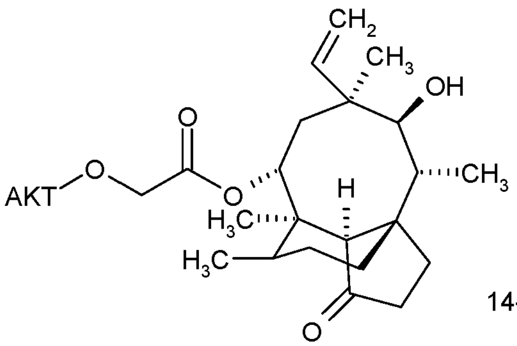
preferably trifluoroacetyl or *tert*-butyloxycarbonyl (BOC).



IIIb

,

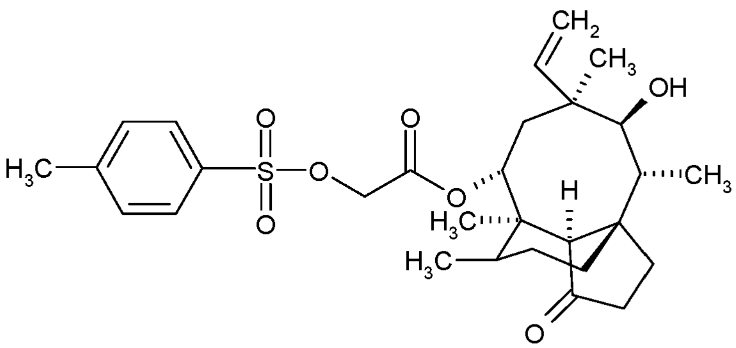
respectively
wherein R is as defined above,
with an activated 14-O-AKT-acetyl-mutilin of formula



14-O-AKT-acetyl-mutilin

,

wherein AKT is an activating
group, optionally mesyl, besyl, tosyl, or—O-AKT is halogen; optionally 14-O-AKT-acetyl-mutilin is a compound of formula



Tos-PLEU

and isolating either a compound of formula IIa, or mixture of a compound of formula IIa with a compound of formula IIb obtained from the reaction mixture.

[0057] Disclosed herein is a compound of formula IIIa in a single stereoisomeric form.

[0058] In a compound of formula IIIa the carbon atoms of the cyclohexyl ring to which the hydroxy group, the amine group and the thio group are attached are all in the R configuration and thus a compound of formula IIIa represents an optionally amino protected (1*R*,2*R*,4*R*)-4-amino-2-hydroxy-1-mercapto-cyclohexane; and in a compound of formula IIIb, respectively, the carbon atoms of the cyclohexyl ring to which the hydroxy group, the amine group and the thio group are attached are all in the S configuration and thus a compound of formula IIIb represents an optionally amino protected (1*S*,2*S*,4*S*)-4-amino-2-hydroxy-1-mercapto-cyclohexane.

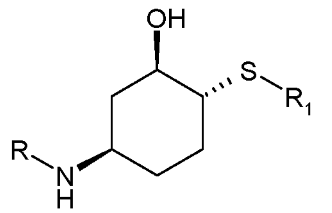
[0059] The coupling reaction of a compound of formula IIIa or the mixture of a compound of formula IIIa with a compound of formula IIIb, respectively, wherein R is as defined above with an activated 14-O-AKT-acetyl-mutilin, wherein AKT is as defined above, to obtain a compound of formula IIa or a mixture of IIa and IIb, wherein R is as defined above may be performed as appropriate, e.g. according, e.g. analogously to a method as conventional, such as under standard conditions known for those reactions; e.g. in the presence of a base, e.g. a strong inorganic base, like a hydroxide such as NaOH, e.g. in a two phase system, and, if the reaction is performed in a two phase system preferably in the presence of a catalyst, such as a phase transfer catalyst, e. g. benzyl-tri-butylammonium chloride. The coupling reaction may be also performed in a single solvent system, e.g. in organic solvent such as a halogenated hydrocarbon, e.g. chlorobenzene, dichloromethane, an aromatic solvent such as toluene, a nitrile such as acetonitrile or an ether, e.g. *tert*-butyl-methylether, tetrahydrofuran, in the presence of base, preferably an organic base, such as DBU.

[0060] Preferably the coupling reaction is performed in organic solvent, e.g. an apolar solvent, such as an ether, e.g. tert-butyl methyl ether (MTBE) or polar solvent, such as a halogenated hydrocarbon, e.g. dichloromethane; preferably in the presence of an aqueous base solution, such as aqueous NaOH; or in the presence of base in the organic solvent, such as DBU, DBN, preferably in the presence of a phase transfer catalyst in case of using an aqueous base solution, such as benzyl-tri-butylammonium chloride.

[0061] It has been found that during the coupling reaction the stereochemistry of the carbon atoms of the cyclohexyl moiety to which the thio, hydroxy and amino group are attached is retained and remains the same as in a compound of formula IIIa and a compound of formula IIIb, respectively, used as a starting material.

[0062] A compound of formula IIIa in the form of a single stereoisomer or the mixture of a compound of formula IIIa with a compound of formula IIIb, respectively, may be prepared as appropriate, e.g. according, e.g. analogously to a method as conventional. Preferably a compound of formula IIIa or the mixture of compound of formula IIIa with a compound of formula IIIb, respectively, are prepared by deprotecting the thiol function in an amino-hydroxy-mercapto cyclohexane, wherein the amino group and the thiol group both are protected.

[0063] In a preferred embodiment, the present invention provides a process according to the present invention, wherein a compound of formula IIIa, or a mixture of a compound of formula IIIa with a compound of formula IIIb is obtained by deprotecting the thiol function either in a compound of formula IVa

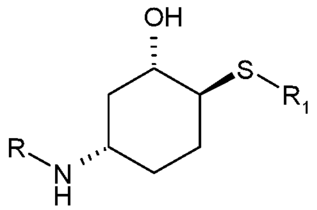


IVa

,

or

in a mixture of compound IVa with a compound of formula IVb



IVb

,

respectively,

wherein R is as defined above and R_1 is a thiol protecting group, and isolating either a compound of formula IIIa, or a mixture of a compound of formula IIIa with a compound of formula IIIb, respectively, obtained from the reaction mixture.

[0064] A thiol protecting group, e.g. in the meaning of R_1 in a compound of formulae IVa and IVb, includes e.g.

- (C_{1-6})alkyl, wherein alkyl optionally is further substituted, e.g. further substiuted by (C_{6-12})aryl such as phenyl, such as a trityl; e.g. removable by strong acid or $AgNO_3$ treatment,
- —(C_{1-6})alkylcarbonyl, e.g. acetyl, e.g. removable by base, such as sodium methoxide, treatment,
- (C_{6-12})arylcarbonyl, such as a benzoyl, e.g. removable by treatment with reduction agent, such as DIBAL, or by treatment with a base, such as hydrazine,

preferably —C(=O)-(C_{6-12})aryl; more preferably benzoyl.

[0065] Appropriate sulfur protecting groups e.g. are described in T. W. Greene, P. G. M. Wuts, Protective Groups in Organic Synthesis, Wiley-Interscience, 4th Edition, 2007, particularly p. 647-695.

[0066] In a compound of formula IVa the carbon atoms of the cyclohexyl ring to which the hydroxy group, the amine group and the thio group are attached are all in the R configuration and thus a compound of formula IVa represents an amine and thio protected (1*R*,2*R*,4*R*)-4-amino-2-hydroxy-1-mercapto-cyclohexane; and in a compound of formula IVb the carbon atoms of the cyclohexyl ring to which the hydroxy group, the amine group and the thio group are attached

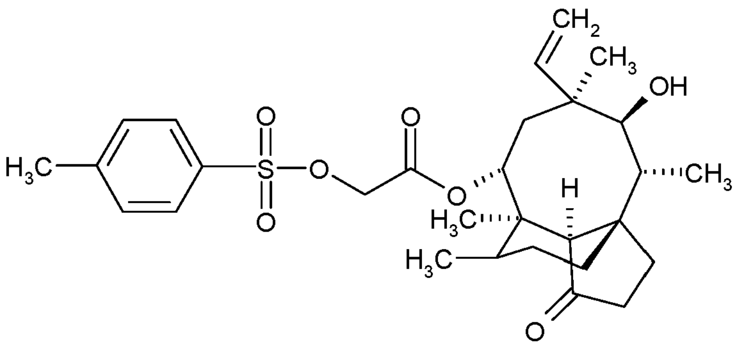
DBN	1,5-diazabicyclo[4.3.0]non-5-ene
DBU	1,8-diazabicyclo[5.4.0]undec-7-ene
DIPE	diisopropylether
DMF	N,N-dimethylformamide
DMSO	dimethylsulfoxide
DMTF	dimethylthioformamide
DPPA	diphenylphosphoryl azide
DTT	1,4-dithio-DL-threitol
ESI	electrospray ionization
EtOAc	ethyl acetate
GF	glass fibre
h	hours
heptane	*n*-heptane
HPLC	High Performance Liquid Chromatography
KF	Karl Fischer
M	molarity
mCPBA	metachloroperoxybenzoic acid
MTBE	methyl tert-butyl ether
min	minutes
MS	mass spectrometry
m/z	mass/charge ratio
t-BuOH	tert-butylalcohol
Bu_4NCl	tetra-*n*-butylammonium chloride
PhCOSH	thiobenzoic acid
rt	room temperature
TLC	thin layer chromatography
TEA, Et_3N	triethylamine
TFA	trifluoroacetic acid
THF	tetrahydrofuran
Wt	weight
w/w	weight/weight
XRPD	powder X-ray diffraction

[0090] A "strip weight assay" as indicated in the examples is defined as follows: The content of an aliquot of a batch or of the whole batch is determined by removing the solvent and determining the content by HPLC or NMR using an internal or external standard and/or subtracting known impurities from the compound. In case of taking an aliquot a back extrapolation to the total mass/volume is performed.

[0091] A "line rinse" as indicated in the examples is a system rinse using an appropriate solvent to minimize losses of product and input materials.

Tos-PLEU is a compound of formula

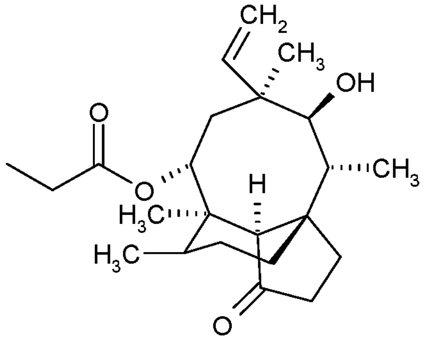
[0092]



Tos-PLEU

PLEU is a residue of formula

[0093]

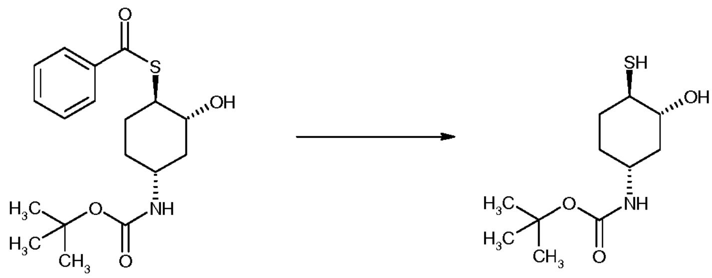


[0094] In the following examples all temperatures are in degrees Celsius (°C) and are uncorrected. The abbreviations used are indicated above (after reaction SCHEME 2).

Example 1

***tert*-Butyl [(1*R*,3*R*,4*R*)-3-hydroxy-4-mercapto-cyclohexyl]-carbamate**

[0095]



[0096] 3.94 Kg of {(1*R*,2*R*,4*R*)-4-[(*tert*-Butoxycarbonyl)-amino]-2-hydroxy-cyclohexyl}-benzene-carbothioate and 37 L of CH_2Cl_2 were charged to a vessel and the mixture obtained was stirred at 15-25°C. 0.39 Kg of 1,4-dithio-DL-threitol (10% wt) was added to the mixture and rinsed through with 2 L of CH_2Cl_2. To the mixture obtained 0.84 Kg of hydrazine monohydrate was added. The mixture obtained was stirred at 18 to 22°C for 3 h and the reaction was followed by HPLC. Upon completion of the reaction, 39 L of 1 M phosphoric acid solution was added and the mixture obtained was stirred for a further 15-30 min. Two phases formed were separated and the organic phase obtained was washed with 39 L of of 1 M phosphoric acid solution followed by 39 L 1% aqueous NaCl solution. The organic layer obtained was concentrated *in vacuo* at <40°C, to the concentration residue 20 L of CH_2Cl_2 was added and the mixture obtained again was concentrated. To the concentration residue obtained a further 8 L of CH_2Cl_2 was added and the mixture obtained was concentrated to dryness.

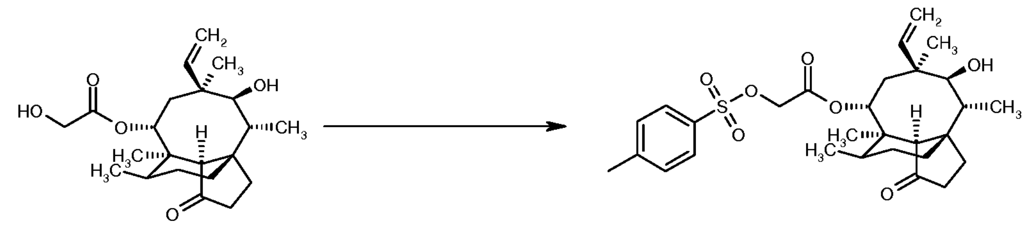
[0097] 2.89 Kg of *tert*-Butyl [(1*R*,3*R*,4*R*)-3-hydroxy-4-mercapto-cyclohexyl]-carbamate in the form of a white solid was obtained.

[0098] ^{1}H NMR (200 MHz, DMSO-d6, ppm) δ 6.79 (d, J=7.8Hz, 1H), 4.99 (d, J=5.8Hz, 1H), 3.34 — 3.24 (m, 1H), 3.14 — 3.04 (m, 1H), 2.37 (d, J=3.8 Hz, 1H), 2.00 -1.89 (m, 1H), 1.87 — 1.82 (m, 1H), 1.73 — 1.67 (m, 1H), 1.47 — 1.04 (m, 12H)

Example 2

22-*O*-Tosylpleuromutilin

[0099]



[0100] 22-O-Tosylpleuromutilin is a known compound from literature. However a preparation procedure is outlined below.

[0101] A solution of 13.0 kg of pleuromutilin and 6.57 kg of 4-toluenesulfonyl chloride in 42.1 L of CH_2Cl_2 at 10 to 15 °C was treated with 9.1 L of 5.7 M aqueous NaOH over 20 min, maintaining a temperature < 25 °C. The resulting off-white suspension was heated to reflux for 20 h and the reaction was followed until completion determined by HPLC. Upon reaction completion the mixture obtained was cooled to 20 to 30 °C, diluted with 52 L of CH_2Cl_2, stirred at 15 to 25 °C for 10 min, and the layers obtained were separated. The organic phase obtained was washed several times with 52 L of water until a pH of the aqueous layer was adjusted to < 9. The organic layer obtained was concentrated to 4 volumes and azeotropically dried twice with 52 L of CH_2Cl_2. To the solution obtained 52 L of heptane were added dropwise and the solution obtained was concentrated at < 40 °C to approximately 4 volumes. To the concentrate obtained 52 L of heptane was added and the resulting suspension was stirred at 20 to 25 °C for 2 to 2.5 h, filtered, the filter cake obtained was washed with 39 L of heptane and pulled dry on the filter.

[0102] The solid was dried under vacuum at < 40 °C for at least 12 h.

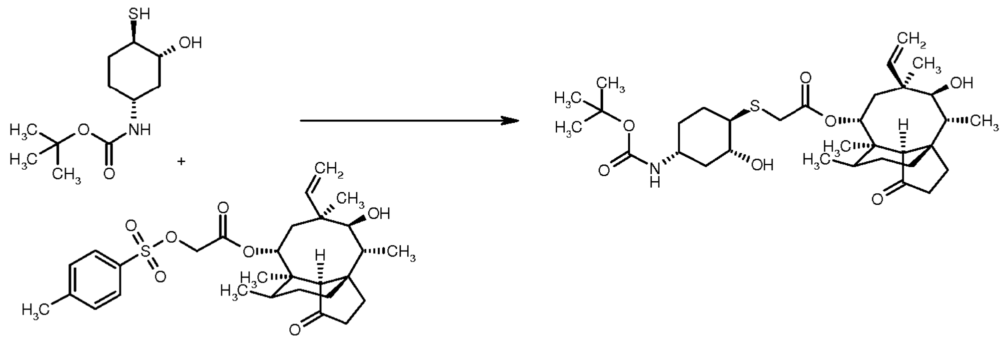
[0103] 16.9 kg of 22-*O*-tosylpleuromutilin in the form of a white solid was obtained.

[0104] ^1H NMR (200 MHz, DMSO-d6, ppm, *inter alia*) δ 7.81 (d, 2H), 7.47 (d, 2H), 6.14 — 6.0 (m, 1H), 5.54 (d, J=7.8Hz, 1H), 5.08 — 4.99 (m, 2H), 4.70 (AB, J=16.2Hz, 2H), 3.41 (d, J=5.2Hz, 1H), 2.41(s, 4H), 1.04(s, 3H), 0.81 (d, 3H), 0.51 (d, 3H)

Example 3

14-*O*-1[(1*R*,2*R*,4*R*)-4-*tert*-Butoxycarbonylamino-2-hydroxy-cyclohexyl-sulfanyl]-acetyl}-mutilin

[0105]



[0106] 4.75 Kg of Pleuromutilin tosylate (Tos-PLEU) and 44.4 L of MTBE were charged into a vessel and to the mixture obtained 0.31 Kg of benzyl-tri-*n*-butylammonium chloride was added and rinsed through with 2.4 L of MTBE. To the mixture obtained 20 L of 1M aqueous NaOH solution and 2.84 Kg of *tert*-Butyl [(1*R*,3*R*,4*R*)-3-hydroxy-4-mercapto-cyclohexyl]-carbamate were added and the mixture obtained was stirred at 17 to 23°C for 3 h. Upon completion of the reaction (determined by HPLC) two layers formed were separated and the lower aqueous layer was removed. The organic phase obtained was washed with 19 L of 1M aqueous NaOH solution, twice with 20 L of 0.1 M phosphoric acid, 20 L of 10% aqueous $NaHCO_3$ solution and twice with 20 L of water. The organic liquors obtained were concentrated, the concentrate obtained was taken up in 7.46 Kg of 2-propanol, the mixture obtained was concentrated again and dried *in vacuo* at <40°C. 6.66 Kg of 14-*O*-{[(1*R*,2*R*,4*R*)-4-*tert*-Butoxycarbonylamino-2-hydroxy-cyclohexyl-sulfanyl]-acetyl} -mutilin in the form of a white foam was obtained.

[0107] ^1H NMR (200 MHz, DMSO-d6, ppm, *inter alia*) δ 6.78 (d, J=7.8Hz, 1H), 6.22 — 6.08 (m,1H), 5.55 (d, J=7.8Hz, 1H), 5.13 — 5.02 (m, 2H), 4.95 (d, J=5Hz, 1H), 4.52 (d, J=6Hz, 1H), 3.36 (AB, J=15Hz, 2H), 2.40 (s, broad, 1H), 2.15 — 2.0 (m, 3H), 1.9 — 1.8 (m, 1H), 1.35 (s, 9H), 0.81 (d, J=7Hz, 3H), 0.62 (d, J=6.6Hz, 3H)